

Railway Timetabling Benchmark Lab: Size-Ladder Report (B0–B6 + P1)

ASU Trans+AI Lab

repo: `asu-trans-ai-lab/railway-timetabling-lab` — `benchmark.lab/`

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Abstract

We report the first complete run of the classical railway scheduling laboratory over a size ladder of single-track (C1) corridor instances plus a double-track (C2) probe. The solver ladder comprises B0 priority heuristics, B1 Lagrangian relaxation, B2 time-indexed MILP (HiGHS), B3 LR-based branch-and-bound, B4 CP-SAT (OR-Tools), B5 individual-path column generation with exact pricing, B6 branch-and-price with departure-window branching, and P1 group-supercolumn CG. All methods emit one common reporting record; every bound is validity-checked against generator ground truth. Headlines: the exact-equivalence gate passes with three independent exact solvers agreeing on every proven instance; the individual-path LP shows a real 18.1% integrality gap that a full-component group supercolumn closes to zero; branch-and-price reaches [26, 29] but cannot certify seed1 with departure-window branching alone (12 departure-inseparable subtrees) — precedence branching is required; instances with 8–12 trains remain open with certified intervals. Two claims made earlier in development were retracted when convergence checks failed — the retractions are part of this report.

1 Instances and ground truth

Feasible-first generation (infrastructure $\rightarrow$ serial-schedule incumbent $\rightarrow$ free-flow intended arrivals $\rightarrow$ release compression), manifest-recorded. Objective: $\sum_k |\text{arr}_k - \text{intended}_k|$; unit capacity per (link, minute) cell with 3-min headway; dwell only on sidings.

Table 1: The size ladder. Proven optima certified by ≥ 2 independent exact solvers.

instance	trains	serial UB	ground truth	status
toy_native	3	—	12	proven (MILP=B&B=CP-SAT=B&P)
C1.L4.n4.seed1	4	75	29	proven (4 solvers)
C1.L4.n4.seed2	4	71	20	proven (4 solvers)
C1.L6.n6.seed3	6	276	57	proven (CP-SAT 0.7s; LR-B&B incumbent agrees)
C1.L8.n8.seed11	8	678	[105, 138]	OPEN
C1.L10.n10.seed12	10	1375	[106, 189]	OPEN
C1.L12.n12.seed13	12	2446	[155, 388]	OPEN
C2.L6.n8.seed21 (double track)	8	544	49	proven (0.1 s, after direction fix)

Table 2: Exact tier. UB = returned objective; proven = optimality certificate. B2 is a documented RESTRICTED model (toward-destination arcs, departure pad, dwell cap): UB-only, no LB claim.

instance	method	UB	LB	proven	time (s)
toy	B0 (any rule)	12	—	no	0.1
toy	B2 MILP	12	—	(restricted)	0.0
toy	B4 CP-SAT	12	12	yes	0.0
toy	B5 CG	12	12.00	LP tight	0.0
toy	B6 B&P	12	12	yes (root)	0.0
seed1	B0 best (class)	29	—	no	0.4
seed1	B2 MILP	29	—	(restricted)	4.6
seed1	B3 LR-B&B	29	29	yes (1385 nodes)	55.7
seed1	B4 CP-SAT	29	29	yes	0.0
seed1	B5 CG	42	23.75	LP converged	4.5
seed1	B6 B&P	29	26	no (12 stalled subtrees)	164.9
seed2	B4 CP-SAT	20	20	yes	0.0
seed2	B5 CG / B6 B&P	20	20.00	LP tight / root	3–4
seed3	B0 best (rand5)	75	—	no	3.9
seed3	B2 MILP	63	—	(restriction binds!)	38.0
seed3	B4 CP-SAT	57	57	yes	0.7
seed3	B5 CG	69	51.00	LP converged	9.0
seed3	B6 B&P	57	51.00	no (budget)	150.9

2 Proven tier: method comparison

Observations. (i) CP-SAT is the fastest exact solver throughout. (ii) On seed3, B2’s dwell cap ($s_{\max} = 15$) binds: its restricted optimum 63 exceeds the true optimum 57 — visible only because the restriction is labeled UB-only. (iii) The LR-based B&B (B3) proves seed1 but on seed3 its Lagrangian bound stalls (36.3 after 6441 nodes) while its incumbent finds the optimum: LR bound weakness quantified per instance (LR 36.3 < path-LP 51 < opt 57).

3 The convexification result (P1)

relaxation (seed1)	LP value	integrality gap	cost
individual paths (B5, exact pricing, converged)	23.75	18.1%	4.5 s
group supercolumns, component split [1, 3]	25.25	12.9%	4.5 s
full 4-train component as one group	29.00 = integer = optimum	0%	114.7 s

Making the strong interaction primal-visible inside one jointly-feasible column removes the integrality gap entirely (single column, one pricing round); the group-size cost (24 order enumerations) is the $K_{\max}$ trade-off. *Label:* the pricer is order-enumeration, not full joint DP, so LP values are *restricted* group LPs; the certified-true-group-LP claim awaits the designed pair joint DP (PLAN §5b). On seed3, split [3, 3] groups add essentially nothing (restricted group LP $\rightarrow$ 51.17 vs individual 51): **component-covering groups are what matter.**

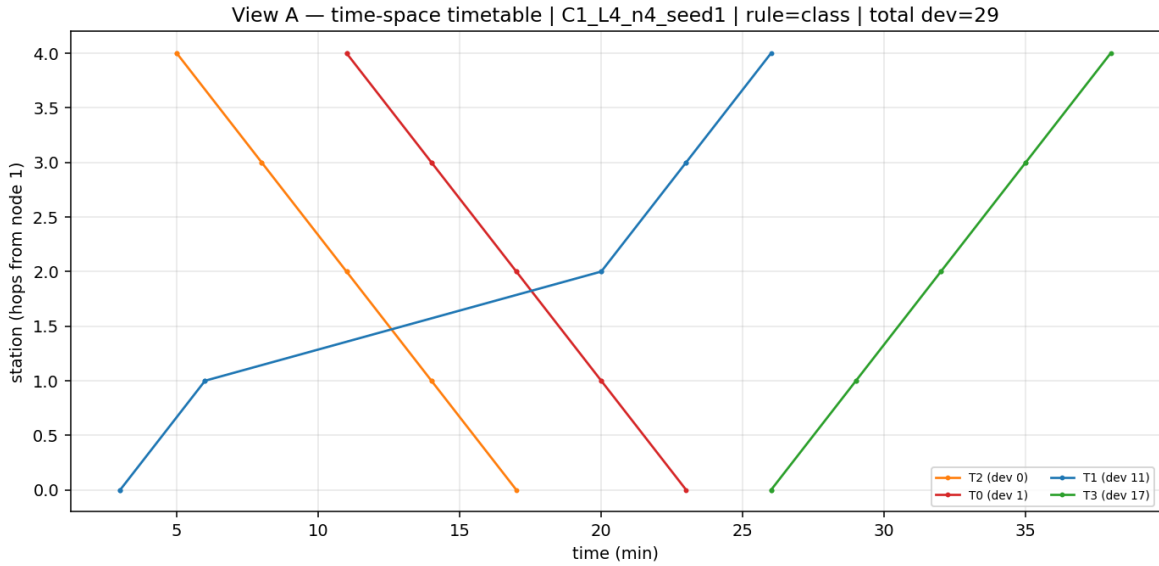
4 Open tier and retractions

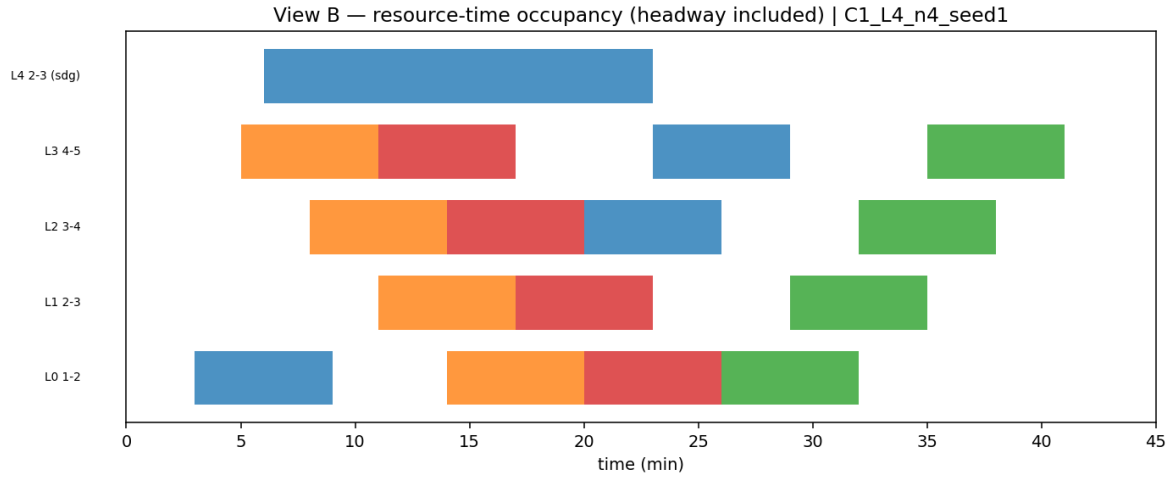
For $n \in \{8, 10, 12\}$ no method proves optimality in budget. Certified intervals come from CP-SAT UBs and, for $n = 8$, a *converged* B&P root LP ($105 > \text{CP-SAT's } 102$). Two development-time claims were retracted by convergence checks and are recorded in the ledger: (a) three B&P node bounds on the open tier (node CG iteration-capped $\Rightarrow$ not valid bounds); (b) the seed1 B&P “proof” (stalled subtrees had been discarded; honest result [26, 29], unproven); (c) the seed3 “75% gap closed” group-LP reading (an unconverged restricted value, 55.50, that fell to 51.17 under deeper pricing). Restricted-master values are meaningful only at pricing convergence.

5 Topology probe (C2)

Eight trains on the same corridor length: single track [105, 138] vs double track **49 (proven)** (an earlier interval [39, 43] was INVALID: the CP-SAT model could traverse one-way tracks backward; fixed and re-proven) — the interaction-density lever the C2/CM experiment family will sweep.

6 Timetable views (seed1, proven optimum)





Reproducibility

`benchmark_lab/run_pipeline.py` regenerates `pipeline_records.csv`; `make_report.py` rebuilds the markdown ledger; every instance carries a `manifest.json` with seed, serial schedule, and ground truth. Common record schema: `record.schema.py`.